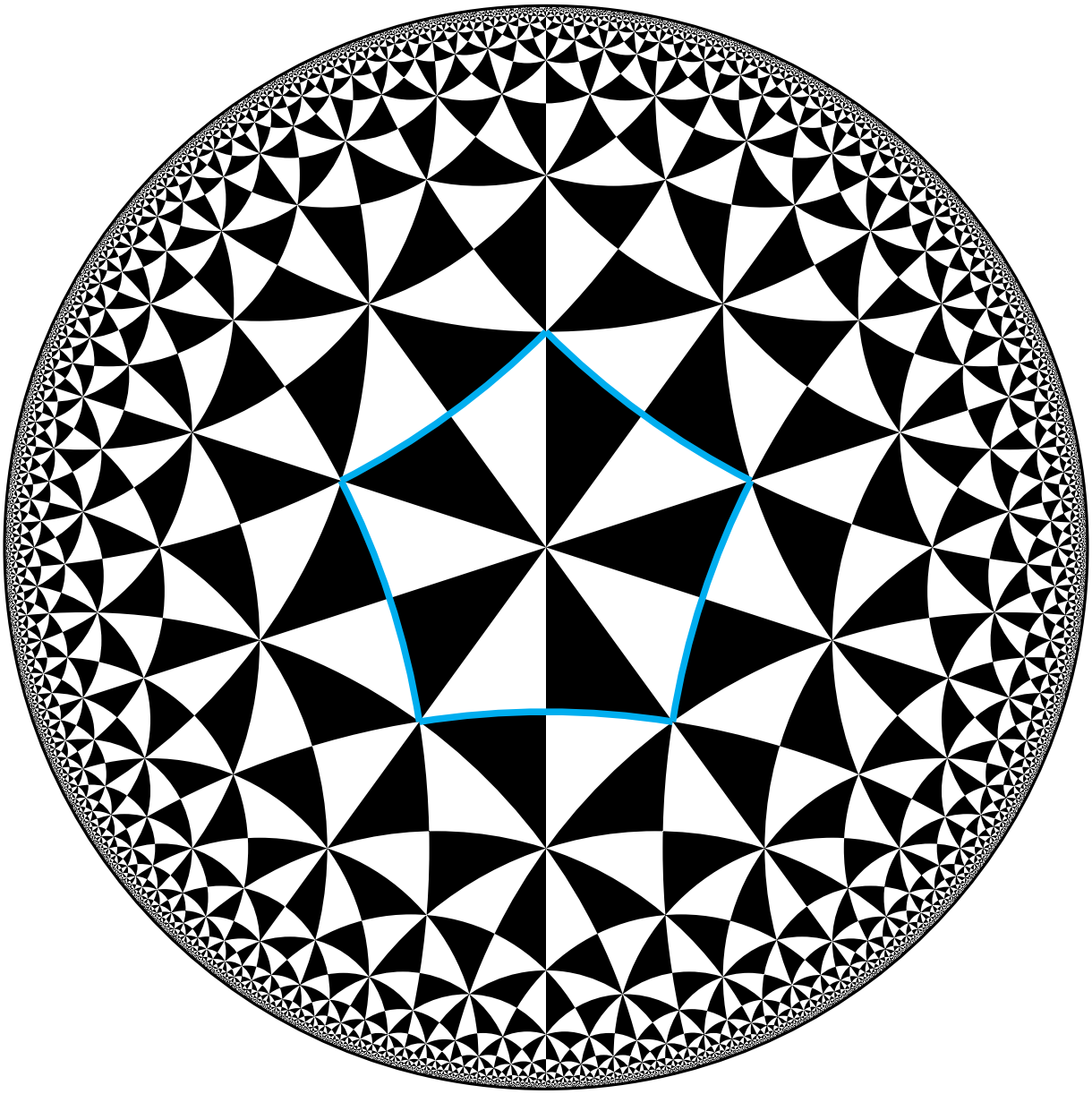
# Package luahyperbolic

Damien Mégy

February 28, 2026



## Contents

1 Introduction 2
1.1 What you can do with this package . . . 2
1.2 What this package cannot do for you . . . 2
1.3 How to load the package . . . 2
1.4 How to customize the package . . . 3
1.5 About the code and pictures shown in this manual . . . 3

2 Drawing elements 3
2.1 Points . . . 3
2.2 Geodesic segments, lines and rays . . . 4
2.3 Circles, semicircles, arcs . . . 7
2.4 Horocycles . . . 9

3 Defining points 9
3.1 Midpoint, interpolate, barycenter . . . 9
3.2 Triangle points (CAUTION) . . . 10
3.3 Intersections . . . 11
3.4 Reflections and projections . . . 12
3.5 (Elliptic) Rotations . . . 13
3.6 Orbits under automorphisms . . . 14

4 Tilings 15

5 Lua libraries 16
5.1 complex.lua . . . 16
5.1.1 Functions . . . 16
5.1.2 Constants (numbers and complex numbers) . . . 16
5.2 luahyperbolic-tikz.lua and hyperbolic-core . . . 17
5.2.1 Functions . . . 17
5.2.2 Constants (numbers and complex numbers) . . . 17

# 1 Introduction

## 1.1 What you can do with this package

:

1. Draw hyperbolic geometry pictures, high precision (pdf output), tiny size on disk
2. ...that you can later convert to svg or raster to bitmap images, if you want.
3. Easy (?) syntax. At least, easier than with LaTeXmacros.
4. Easy to read, educational Lua source code if you want to learn hyperbolic geometry.
5. No need to install Python packages or create an account on some webapp.
6. Compile anywhere in a single pass with lualatex compiler, available everywhere. No (Leeloo Dallas) multipass.
7. Doesn't need to compile in "shell-escape" mode.

## 1.2 What this package cannot do for you

1. Intensive computations. Source code is not optimized to keep maximum readability. All computations involve automorphisms. For heavy computations, please use a ~~Python~~ C library. The above (embedded) image compiled in approx. 10 seconds on a 2020 intel macbook air : 1.7 seconds to compute the 3783 geodesics wit Lua, and 8 seconds to draw them with TikZ. This documentation compiles in 10s including all the tilings in last section.
2. Coffee

## 1.3 How to load the package

Local install

Simply put the `luahyperbolic.sty` file in your working directory and add the following line to your tex preamble :

```
\usepackage{luahyperbolic}
```

The file `luahyperbolic.sty` contains everything needed, including the necessary library for complex numbers. It can be sent to colleagues by email etc.

It will declare two lua globals : `complex` and `hyper`, and define a `hyperbolicdisk` environment for convenience.

If the sty file is not in the same directory as your tex file, you can write `\usepackage{../luahyperbolic}`

CTAN

: not yet available

### 1.4 How to customize the package

The `luahyperbolic.sty` is generated from three lua modules : `complex.lua`, `luahyperbolic-core.lua` and `luahyperbolic-tikz.lua`. These files are in the `lua/` subdirectory of the package repo.

To build the `luahyperbolic.sty` from lua source files, execute `lua build.lua` in the root directory of the package. This will backup the old sty file to the `backup/` directory, then buid and replace the existing sty file.

### 1.5 About the code and pictures shown in this manual

In this documentation, we use the shortcut environment `\begin{hyperbolicdisk} ... \end{hyperbolicdisk}` which is defined in the package as follows:

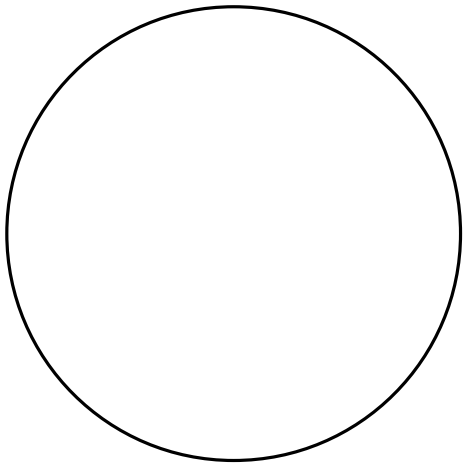
```
\newenvironment{hyperbolicdisk}[1][scale=3]{%
  \begin{tikzpicture}[#1]
    \draw[very thick] (0,0) circle (1);
    \clip (0,0) circle (1);
}{%
  \end{tikzpicture}
}
```

## 2 Drawing elements

### 2.1 Points

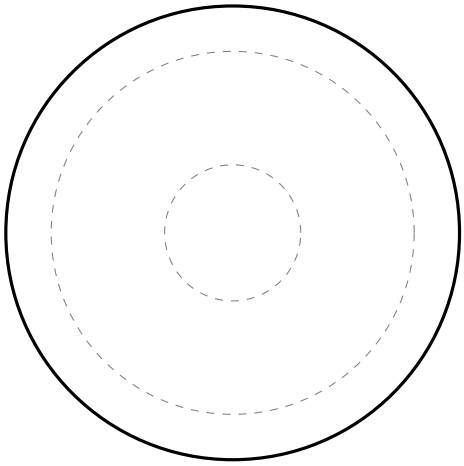
Draw and label points

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
local D = complex(-0.6,0.2)
hyper.drawPoint(A)
hyper.drawPoints(B, C, "teal")
hyper.drawPoint(D, "white, draw=red")
hyper.labelPoint(A,"$A$") -- default : above
    ↪ left
hyper.labelPoints(B,"$B$", C, "$C$", "above
    ↪ right, teal")
hyper.labelPoint(D,"$D$","below=.4cm, red")
-- hyper.drawPoint(complex(2,2)) -- will error
\end{luacode*}
\end{hyperbolicdisk}
```



Random points with uniform density

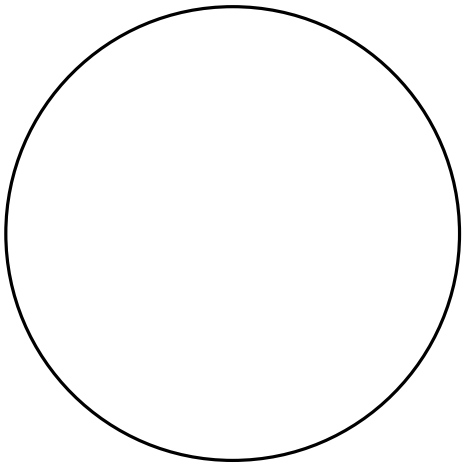
```
\begin{hyperbolicdisk}
\begin{luacode*}
for k=1,100 do
  local M = hyper.randomPoint(0.3,0.8)
  hyper.drawPoint(M)
end
\end{luacode*}
\draw[gray, dashed] (0,0) circle(0.3) circle
    ↪ (0.8);
\end{hyperbolicdisk}
```



## 2.2 Geodesic segments, lines and rays

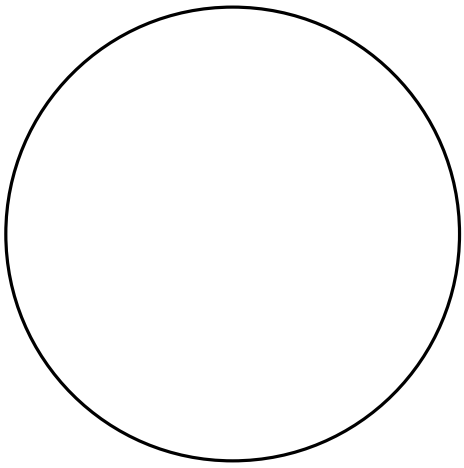
Geodesic segments

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.2,0.1)
local B = complex(-0.3,0.5)
local C = complex(0.6,-0.5)
hyper.drawSegment(A, B, "thick,blue")
hyper.drawSegment(A, C, "dashed,ForestGreen")
hyper.drawPoints(A,B,C)
\end{luacode*}
\end{hyperbolicdisk}
```



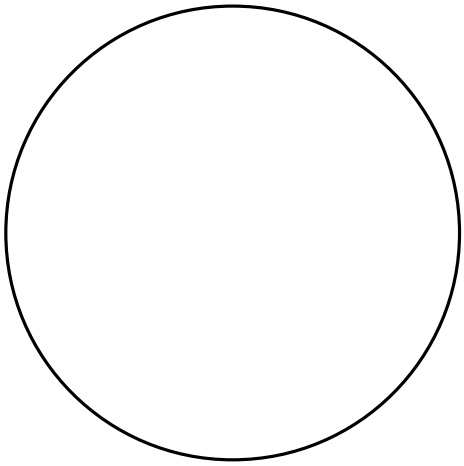
Lines and rays (through two points)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(0.5,0.6)
local C = complex(-0.5,0.5)
local D = complex(-0.7,-0.3)
hyper.drawLine(A, B, "red")
hyper.drawRay(C,D, "blue")
hyper.drawLine(D,-D, "green!50!black")
hyper.drawLine(A,complex(1,0), "gray")
hyper.drawLine(complex(1,0), complex(0,1), "
    ↪ teal")
hyper.drawPoints(A, B, C, D)
hyper.labelPoints(A, "$A$", B, "$B$", C, "$C$",
    ↪  D, "$D$")
\end{luacode*}
\end{hyperbolicdisk}
```



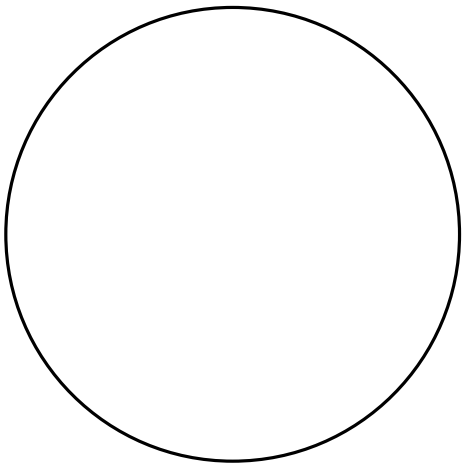
## Triangles and polygons

```
\begin{hyperbolicdisk}
\begin{luacode*}
hyper.drawTriangle(
  complex(.9,.2),
  complex(.5,.7),
  complex(.7,-.2), "red")
local points={}
for k=1,5 do
  table.insert(points, complex.polar(.5,k*2*
      ↪ math.pi/5))
end
hyper.drawPolygonFromTable(points,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



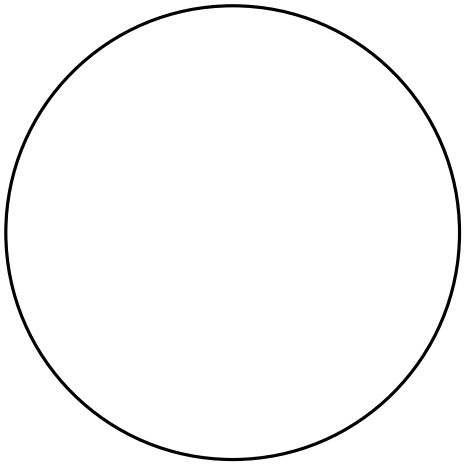
## Regular polygons

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O1 = complex(.5,.2)
local A1 = complex(.7,.3)
local O2 = complex(-0.1,0)
local A2 = complex(-.4,.3)
local O3 = complex(0,.7)
local A3 = complex(0,.9)
hyper.drawRegularPolygon(O1, A1, 3, "red")
hyper.drawRegularPolygon(O2, A2, 5, "blue")
hyper.drawRegularPolygon(O3, A3, 8, "green!50!
    ↪ black")
hyper.drawPoints(O1,O2,O3)
\end{luacode*}
\end{hyperbolicdisk}
```



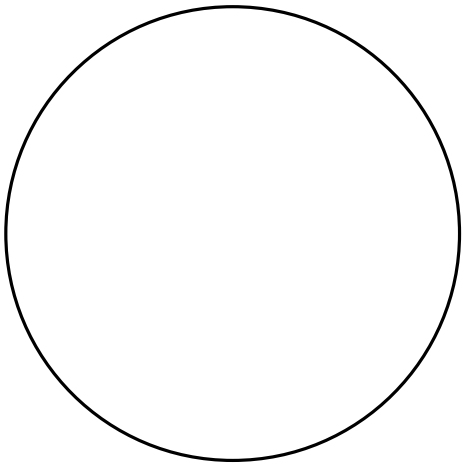
## Polygonal lines

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-.9,.2)
local B = complex(-.5,.8)
local C = complex(-.7,-.4)
local D = complex(.7,-.5)
hyper.drawPolyline(A, B, C, D, "red")
local points={}
for k=1,6 do
  table.insert(points, hyper.randomPoint(0,.7))
end
hyper.drawPolylineFromTable(points,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



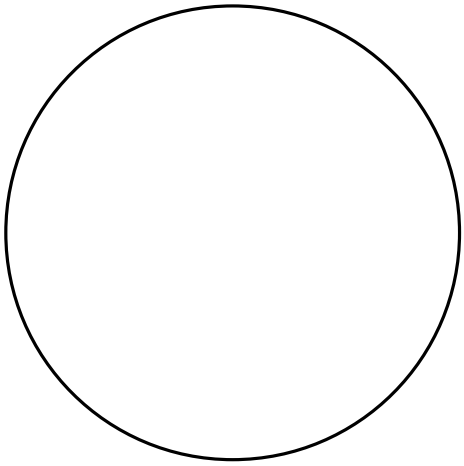
## Lines and rays (point and tangent vector)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(-0.5,0.5)
local C = complex(0,0.2)
local v =  complex(1,0)
hyper.drawLineFromVector(A, v, "red")
hyper.drawRayFromVector(B, v, "blue")
hyper.drawRayFromVector(C, v, "Green")
hyper.drawPoints(A, B,C)
hyper.drawTangentVector(A,v,"red")
hyper.drawTangentVector(B,v,"blue")
hyper.drawTangentVector(C,v,"Green")
\end{luacode*}
\end{hyperbolicdisk}
```



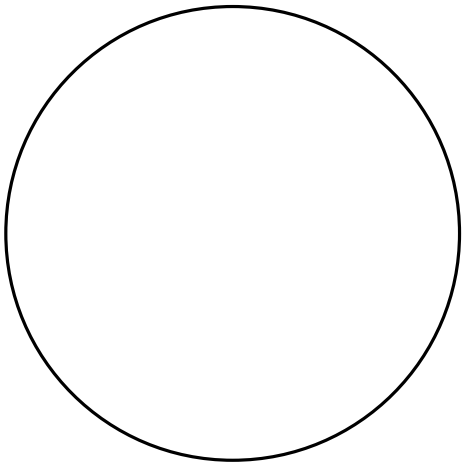
## Perpendicular bisector

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.5)
local B = complex(-0.7,0.3)
hyper.drawPerpendicularBisector(A,B,"red")
hyper.drawPoints(A,B)
hyper.drawSegment(A, B, "teal")
hyper.labelPoints(A,"$A$", B, "$B$", "above
    ↪ right")
\end{luacode*}
\end{hyperbolicdisk}
```



## Angle bisector

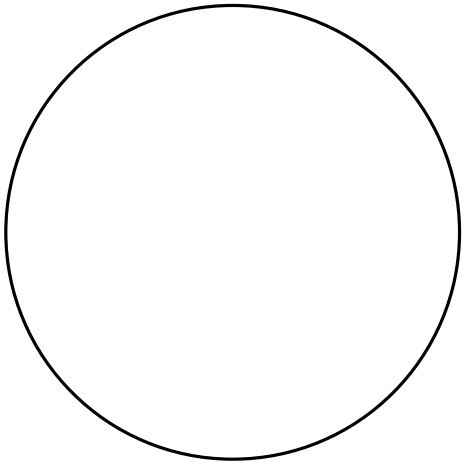
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(-0.2,-0.6)
local O = complex(0.3,0.3)
local B = complex(-0.7,0.3)
hyper.drawAngleBisector(A,O,B,"red")
hyper.drawPoints(A,O,B)
hyper.drawPolyline(A,O,B, "teal")
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
\end{luacode*}
\end{hyperbolicdisk}
```



## 2.3 Circles, semicircles, arcs

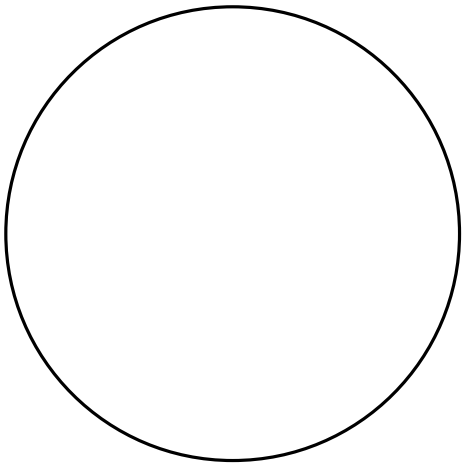
### Center and radius

```
\begin{hyperbolicdisk}
\begin{luacode*}
local points = {
  complex(0,0),
  complex(-0.5,-0.4),
  complex(0.8,0.1),
  complex(0,0.9)
}
for _,point in ipairs(points) do
  hyper.drawCircle(point,1,"teal")
  hyper.drawPoint(point)
end
-- NOTE : all circles have same radius=1 !
\end{luacode*}
\end{hyperbolicdisk}
```



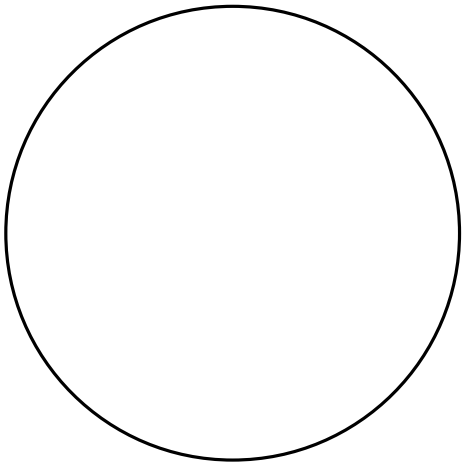
### Circle through point

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local P = complex(0.7,-0.4)
local Q = complex(-0.1,0.4)
hyper.drawCircleThrough(O, P, "red")
-- or by hand :
hyper.drawCircle(O, hyper.distance(O,Q), "blue"
    ↪ )
hyper.drawPoints(O, P, Q)
hyper.labelPoints(O, "$O$", P, "$P$", Q, "$Q$")
\end{luacode*}
\end{hyperbolicdisk}
```



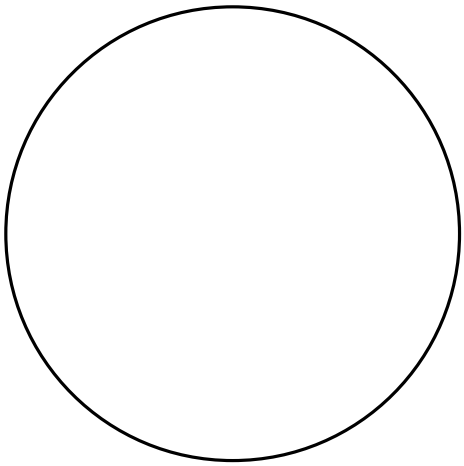
### Semicircles

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0,0)
local A = complex(-0.3,0)
local P = complex(0.5,0.2)
local B = complex(0.7,-0.4)
hyper.drawSemicircle(O, A, "blue")
hyper.drawSemicircle(P, B, "red")
hyper.drawPoints(O, A, P, B)
hyper.labelPoints(O, "$O$", A, "$A$", P, "$P$",
    ↪  B, "$B$")
hyper.drawLine(P,B, "gray, dashed")
\end{luacode*}
\end{hyperbolicdisk}
```



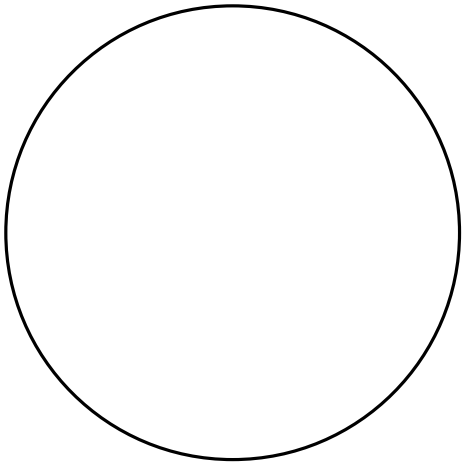
## Arc (center and endpoints)

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0,0)
local A = complex(0.5,0)
local B = complex.polar(0.5,math.pi/3)
hyper.drawArc(O,A,B, "red")
hyper.drawPoints(O,A,B)
\end{luacode*}
\end{hyperbolicdisk}
```



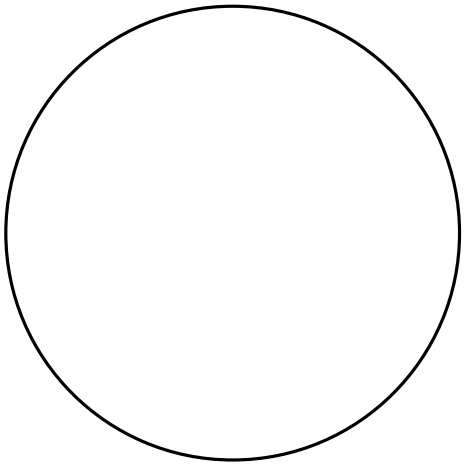
## Draw angle $\widehat{AOB}$

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(-.5,-.3)
local A = complex(0.5,0)
local B = complex(0.2,.7)
hyper.drawLines(O, A, O, B, "teal")
hyper.drawAngle(A,O,B, "red, very thick")
hyper.drawPoints(A,O,B)
hyper.labelPoints(A, "$A$", O, "$O$", B, "$B$")
-- poor man's labelAngle:
hyper.labelPoint(O,"$\\theta$", "above=.4cm,
    ↪ right=.5cm")
\end{luacode*}
\end{hyperbolicdisk}
```



## Tangent at point

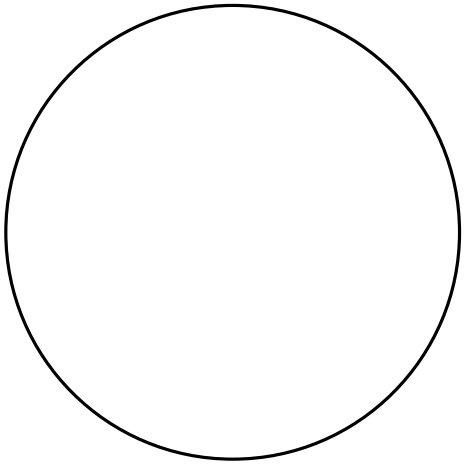
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local P = complex(0.1,-0.4)
local Q = hyper.rotation(O,math.pi/3)(P)
hyper.drawCircleThrough(O, P)
hyper.drawPoints(O, P, Q)
hyper.drawTangentAt(O,P,"red")
hyper.drawTangentAt(O,Q,"teal")
\end{luacode*}
\end{hyperbolicdisk}
```



## 2.4 Horocycles

**Horocycles**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local theta=math.pi/3
local idealPoint = complex.exp_i(theta)
local N = 10
for k=1,N-1 do
  local point = idealPoint * (2*k/N-1)
  hyper.drawHorocycle(idealPoint,point,"red")
  -- and an orthogonal geodesic :
  local endPoint = complex.exp_i(theta+2*k*math
      ↪ .pi/N)
  hyper.drawLine(idealPoint,endPoint,"gray")
end
\end{luacode*}
\end{hyperbolicdisk}
```

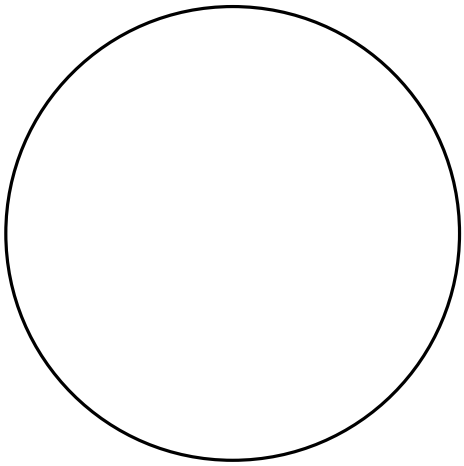


# 3 Defining points

## 3.1 Midpoint, interpolate, barycenter

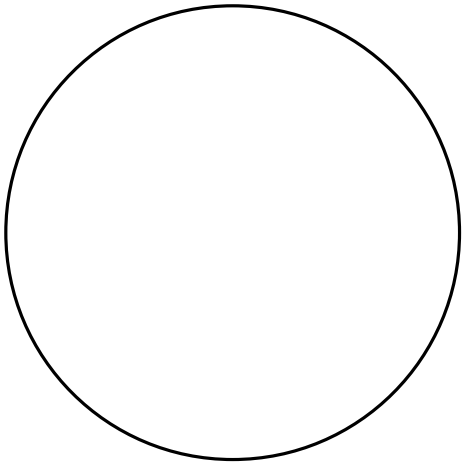
**Midpoint**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.4,0.4)
local B = complex(-0.8,-0.3)
local M = hyper.midpoint(A,B)
hyper.drawSegment(A,B, "teal")
hyper.drawPoints(A,B,M)
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$")
\end{luacode*}
\end{hyperbolicdisk}
```



**Interpolate and barycenter**

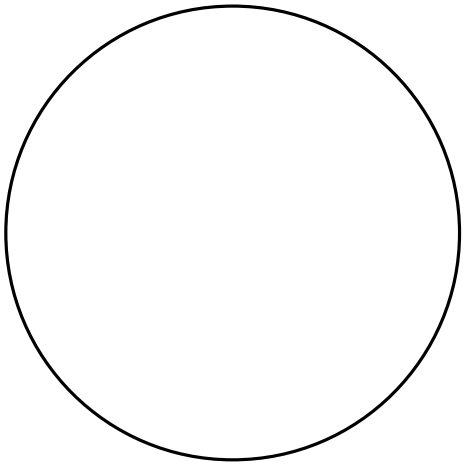
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.3,0.8)
local B = complex(-0.7,-0.4)
hyper.drawSegment(A,B, "teal")
for k=0,10 do
  local M = hyper.interpolate(A, B, k/10)
  hyper.drawPoint(M)
  if k % 2 == 0 then
    hyper.labelPoint(M, "$M_{" .. k .. "}$", "
        ↪ below right")
  end
end
\end{luacode*}
\end{hyperbolicdisk}
```



## 3.2 Triangle points (CAUTION)

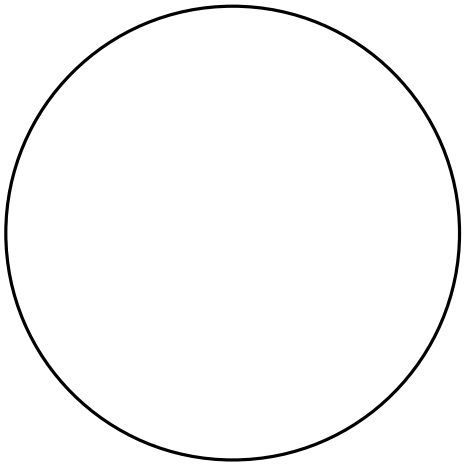
**Centroid (intersection of medians)**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local AA = hyper.midpoint(B,C)
local BB = hyper.midpoint(C,A)
local CC = hyper.midpoint(A,B)
local G = hyper.triangleCentroid(A,B,C)
hyper.drawTriangle(A,B,C)
hyper.drawSegments(A, AA, B, BB, C, CC, "teal")
hyper.drawPoints(AA,BB,CC)
hyper.drawPoint(G,"white, draw=teal")
-- CAUTION :
local wrong = hyper.interpolate(A,AA,2/3)
hyper.drawPoint(wrong, "red")
\end{luacode*}
\end{hyperbolicdisk}
```



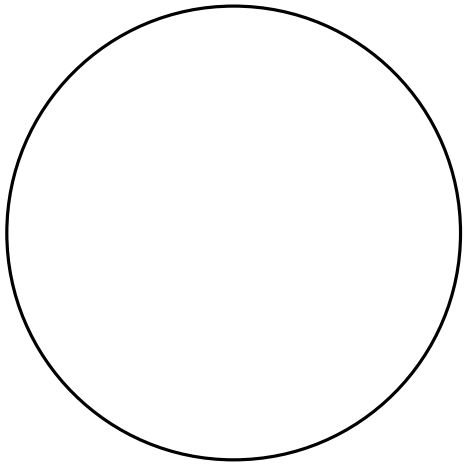
**Circumcenter and circumcircle (CAUTION : does not always exist)**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.4)
local C = complex(0.6,-0.2)
hyper.drawTriangle(A,B,C,"teal")
hyper.drawPerpendicularBisector(A,B, "red")
hyper.drawPerpendicularBisector(B,C, "red")
hyper.drawPerpendicularBisector(C,A, "red")
local O = hyper.circumcenter(A,B,C)
-- Does not always exist !!
if O ~= nil then
  hyper.drawPoint(O,"red")
  local r = hyper.distance(O,A)
  hyper.drawCircle(O,r,"red, dashed")
  hyper.labelPoint(O,"$O$", "below left")
end
\end{luacode*}
\end{hyperbolicdisk}
```



**Incenter and incircle**

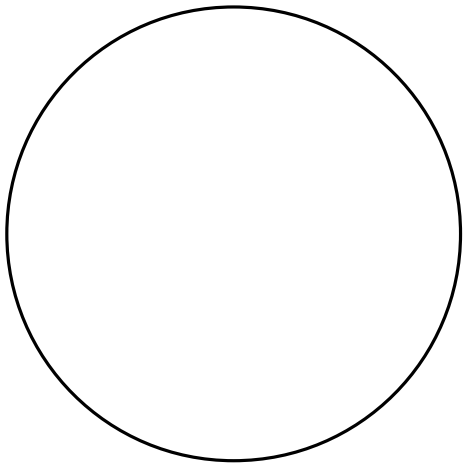
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.1,0.8)
local B = complex(-0.7,-0.3)
local C = complex(0.6,-0.2)
local I = hyper.incenter(A,B,C)
hyper.drawTriangle(A,B,C,"teal")
hyper.drawPoint(I,"red")
hyper.labelPoint(I,"$I$", "below left")
hyper.drawIncircle(A,B,C,"red")
-- show intersection :
hyper.drawAngleBisector(A,B, C, "red!50")
hyper.drawAngleBisector(B,C,A, "red!50")
hyper.drawAngleBisector(C,A,B, "red!50")

\end{luacode*}
\end{hyperbolicdisk}
```



## 3.3 Intersections

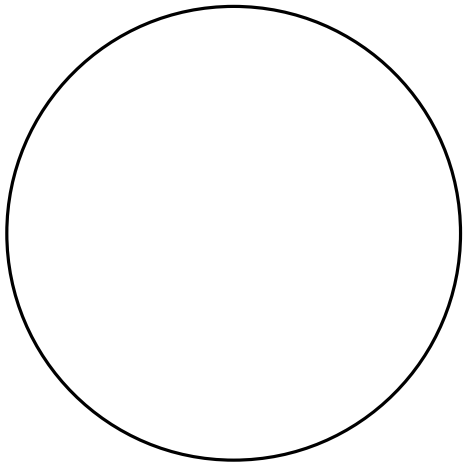
**Line-Line (LL) intersection**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.4,0.3)
local B = complex(-0.4,-0.7)
local C = complex(0,0.7)
local D = complex(0.8,-0.2)
local P = hyper.interLL(A, B, C, D)
hyper.drawLines(A, B, C, D, "teal")
hyper.drawPoints(A, B, C, D,"teal")
hyper.drawPoint(P,"red")
hyper.labelPoint(A, "$A$", "above")
hyper.labelPoint(B, "$B$")
hyper.labelPoint(C, "$C$", "left")
hyper.labelPoint(D, "$D$", "above")
hyper.labelPoint(P, "$P$", "left=.1cm, red")
\end{luacode*}
\end{hyperbolicdisk}
```



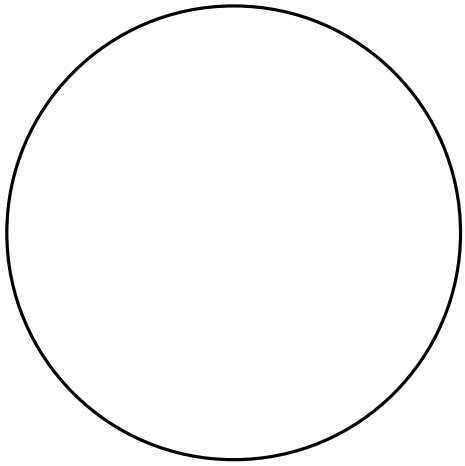
**Line-circle (LC) intersection**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.5,0.4)
local B = complex(-0.3,-0.7)
local O = complex(-0.2,0.1)
local radius = 1
local P, Q = hyper.interLC(A, B, O, radius)
hyper.drawLine(A, B, "teal")
hyper.drawCircle(O,radius, "teal")
hyper.drawPoints(A, B, O, "teal")
hyper.drawPoints(P, Q, "red")
hyper.labelPoints(A, "$A$", B, "$B$", O, "$O$")
hyper.labelPoints(P, "$P$", Q, "$Q$", "red,
    ↪ below right")
\end{luacode*}
\end{hyperbolicdisk}
```



Circle-circle (CC) intersection

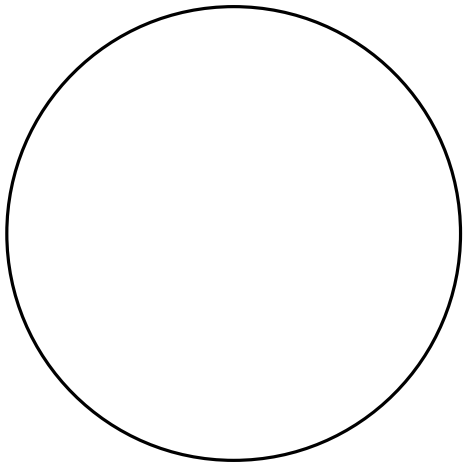
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O1 = complex(-0.6,-.3)
local O2 = complex(0.5,0)
local r1, r2 = 1, 2
local P, Q = hyper.interCC(O1, r1, O2, r2)
hyper.drawCircle(O1,r1, "teal")
hyper.drawCircle(O2,r2, "teal")
hyper.drawPoints(P, Q, "red")
hyper.drawPoints(O1, O2, "teal")
hyper.labelPoints(O1, "$O_1$", O2, "$O_2$", "
    ↪ left, teal")
hyper.labelPoints(P, "$P$", "above left, red")
hyper.labelPoint(Q, "$Q$", "red, below")
\end{luacode*}
\end{hyperbolicdisk}
```



## 3.4 Reflections and projections

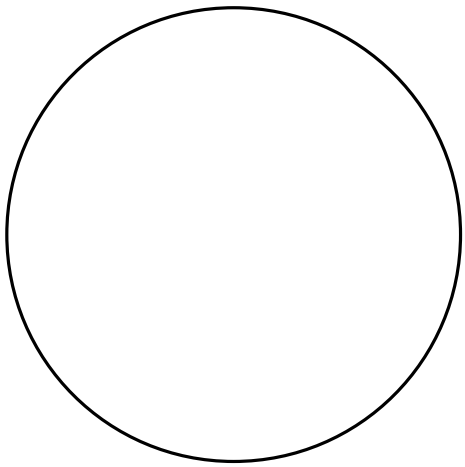
Reflection across a geodesic line

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local Mp = (hyper.reflection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(Mp,"red")
hyper.drawSegment(M,Mp,"dashed,red")
hyper.labelPoint(Mp,"$M'$", "left, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
\end{luacode*}
\end{hyperbolicdisk}
```



Projection onto a geodesic line

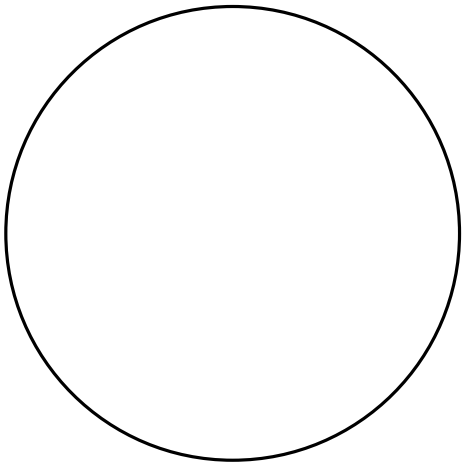
```
\begin{hyperbolicdisk}
\begin{luacode*}
local A = complex(0.8,0.2)
local B = complex(-0.3,0.5)
local M = complex(0.3,-0.6)
local P = (hyper.projection(A,B))(M)
hyper.drawLine(A, B, "teal")
hyper.drawPoints(A, B, M)
hyper.drawPoint(P,"red")
hyper.drawSegment(M,P,"dashed,red")
hyper.labelPoint(P,"$P$", "above, red")
hyper.labelPoints(A, "$A$", B, "$B$", M, "$M$",
    ↪  "below")
\end{luacode*}
\end{hyperbolicdisk}
```



## 3.5 (Elliptic) Rotations

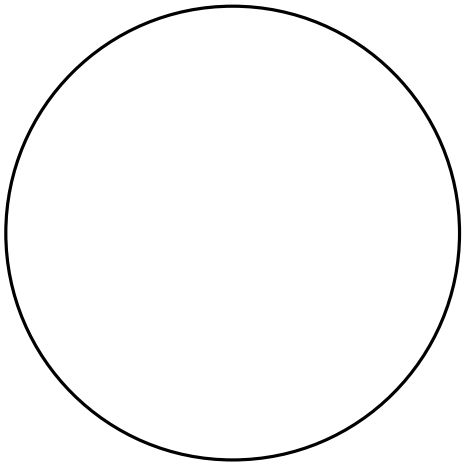
**(Elliptic) rotation around a point (regular octogon)**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.3,0.2)
local M = complex(0.5,-0.4)
hyper.drawPoints(O)
for k=1,8 do
  local P = (hyper.rotation(O,2*math.pi/8))(M)
  hyper.drawSegment(M, P, "red")
  hyper.drawSegment(O,M, "dashed,red")
  M = P
end
hyper.drawCircle(O,hyper.distance(O,M),"thin,
    ↪ gray")
\end{luacode*}
\end{hyperbolicdisk}
```



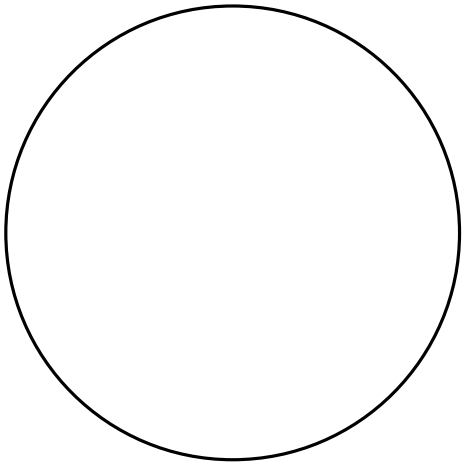
**Symmetry around a point (rotation by 180°)**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.2,0.2)
local s = hyper.symmetry(O)
hyper.drawPoints(O)
local points={complex(-.8,.1), complex(.6,0),
    ↪ complex(0,-.5)}
for k, P in ipairs(points) do
  hyper.drawCircleThrough(O, P, "gray!10")
  local sP = s(P)
  hyper.drawPoints(P,sP)
  hyper.drawSegment(P,sP, "red, dashed")
  hyper.labelPoint(P, "$P_{".. k .."}$")
  hyper.labelPoint(sP, "$\\sigma(P_{".. k .."})
      ↪ $")
end
\end{luacode*}
\end{hyperbolicdisk}
```



**Rotation by 60°**

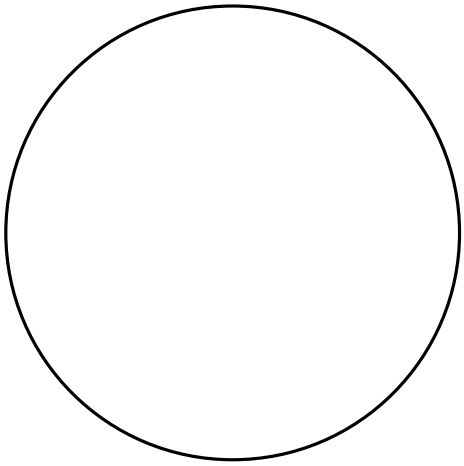
```
\begin{hyperbolicdisk}
\begin{luacode*}
local O = complex(0.1,0.1)
local r = hyper.rotation(O,math.pi/3)
hyper.drawPoints(O)
for k=1,5 do
  local P = hyper.randomPoint(0.3,0.9)
  local rP = r(P)
  hyper.drawCircleThrough(O,P, "gray!10")
  hyper.drawArc(O,P,rP,"red")
  hyper.drawPolyline(P,O,rP, "red!10")
  hyper.drawPoints(P,rP)
end
\end{luacode*}
\end{hyperbolicdisk}
```



## 3.6 Orbits under automorphisms

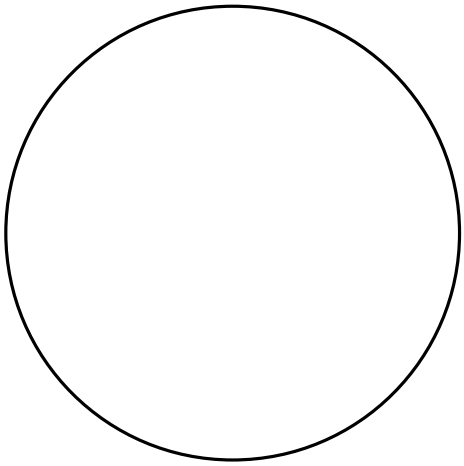
**Orbit under elliptic automorphism (rotation)**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbPoints = 7
local nbIterations = 5
local O = complex(0.1,0.2)
local rot = hyper.rotation(O,math.pi/20) --
    ↪ angle in radians
hyper.drawPoint(O,"red")
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawCircleThrough(O,P,"draw opacity=0.2
      ↪ ")
  hyper.drawPointOrbit(P, rot, nbIterations)
end
\end{luacode*}
\end{hyperbolicdisk}
```



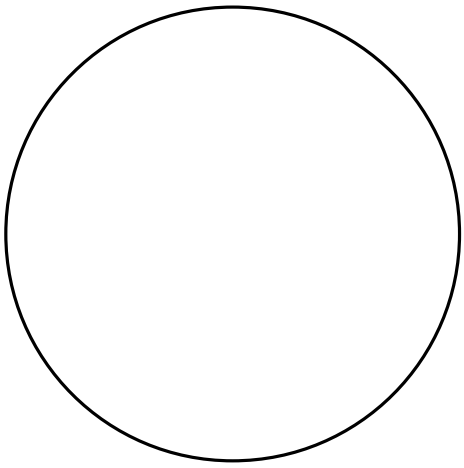
**Orbit under hyperbolic automorphism**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbPoints = 10
local nbIterations = 5
local A = complex(0.1,0.1)
local phi = hyper.automorphism(A,0)
for k=1,nbPoints do
  local P = hyper.randomPoint(0,.9)
  hyper.drawPointOrbit(P, phi, nbIterations)
end
hyper.drawPoint(A,"red")
hyper.labelPoint(A,"$A$", "above left, red")
hyper.drawPointOrbit(A, phi, nbIterations, "red
    ↪ ")
\end{luacode*}
\end{hyperbolicdisk}
```



**Orbit under parabolic automorphism**

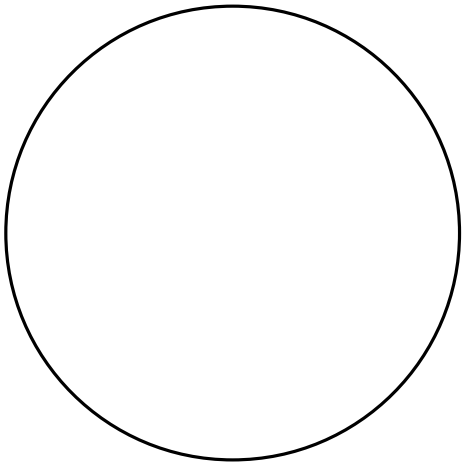
```
\begin{hyperbolicdisk}
\begin{luacode*}
local nbIterations = 7
limit = complex.exp_i(math.pi/3) -- on circle
local phi = hyper.parabolic(limit,math.pi/20)
local points={
  complex(0,0), complex(0,.7),
  complex(.4,.4), complex(.3,.7),
  complex(-.8,.4), complex(-.4,0),
  complex(.4,-.45)}
for _,P in ipairs(points) do
  hyper.drawPointOrbit(P, phi, nbIterations)
  hyper.drawHorocycle(limit, P, "red, draw
      ↪ opacity=.3")
end
\end{luacode*}
\end{hyperbolicdisk}
```



# 4 Tilings

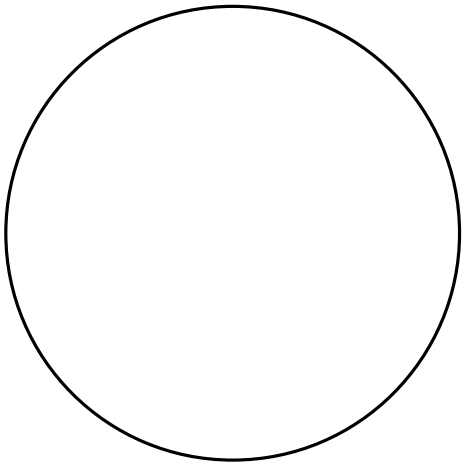
**Triangle tiling**

```
\begin{hyperbolicdisk}
\begin{luacode*}
hyper.SCALE=1e5
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local geodesics = {
{hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



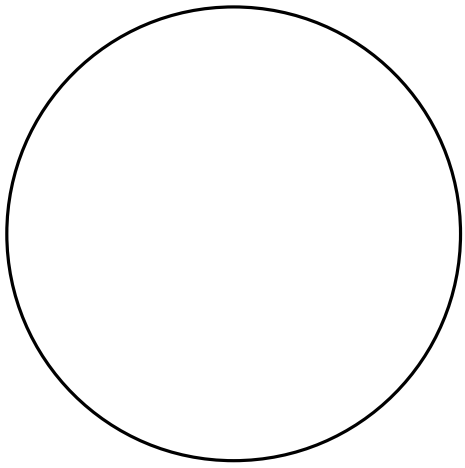
**Same tiling but centered on the pentagon**

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local phi = hyper.automorphism(C,0)
A, B, C = phi(A), phi(B), phi(C)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawRegularPolygon(complex(0,0),B,5, "
    ↪ very thick, green!50!black")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



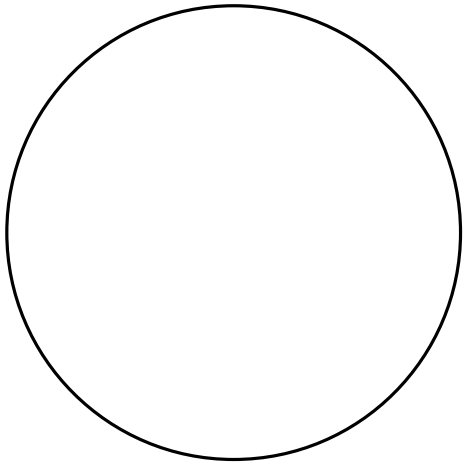
Same tiling but centered on the square

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalRightTriangle
    ↪ (4,5)
local phi = hyper.automorphism(B,0)
A, B, C = phi(A), phi(B), phi(C)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "green!50!
    ↪ black, draw opacity=.5")
hyper.drawRegularPolygon(complex(0,0),C,4, "
    ↪ very thick, green!50!black")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



Ideal triangle tiling

```
\begin{hyperbolicdisk}
\begin{luacode*}
local A, B, C = hyper.fundamentalIdealTriangle
    ↪ (3,2)
local geodesics = {
  {hyper.endpoints(A,B)},
  {hyper.endpoints(B,C)},
  {hyper.endpoints(C,A)}
}
geodesics = hyper.propagate_geodesics(geodesics
    ↪ , TILING_DEPTH)
hyper.drawLinesFromTable(geodesics, "teal, draw
    ↪  opacity=.5")
hyper.drawTriangle(A,B,C,"red, very thick")
\end{luacode*}
\end{hyperbolicdisk}
```



# 5 Lua libraries

## 5.1 complex.lua

### 5.1.1 Functions

| | | | | | |
|---|---|---|---|---|---|
| __add | __unm | det | isClose | isReal | rotate |
| __div | abs | distinct | isClosePair | isUnit | rotate90 |
| __eq | abs2 | dot | isColinear | log | scal |
| __mul | arg | exp | isImag | new | toPolar |
| __sub | clone | exp_i | isNear | nonzero | unit |
| __tostring | conj | invert | isNot | polar | |

### 5.1.2 Constants (numbers and complex numbers)

EPS, EPS_INV, I, J, ONE, ZERO, __index,

## 5.2 luahyperbolic-tikz.lua and hyperbolic-core

### 5.2.1 Functions

### 5.2.2 Constants (numbers and complex numbers)